Literature Review: Deep Learning and AI-Based Cattle Feed and Feeding Monitoring (2026 Studies)

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Ref	Paper Title	Author(s) / Year	Journal	Month & Year	Methodology & Technology	Advantages	Limitations
1	**AI-Based Framework to Predict Animal and Pen Feed Intake in Feedlot Beef Cattle**	Maia, A. S. C., Hall, J. B., Milan, H. F. M. & Teixeira, I. A. M. A. (2026)	Smart Agricultural Technology	2026 (month not confirmed)	Machine learning on 11 years of electronic-feeder (GrowSafe) data from an Idaho feedlot plus weather-station data; two new environmental indices (InComfort-Index, EASI-Index); XGBoost gave the best predictions of daily intake per animal and per pen.	Predicts feed intake at animal and pen level (RMSE 1.38 kg/day per animal; 0.14 kg/day-animal at pen level); accounts for climate; supports feed-waste reduction and feed delivery planning.	Data from a single feedlot facility; depends on electronic feeding systems that many farms lack; InComfort-Index predicted feed intake poorly, and EASI-Index needs feeding-behavior data.
2	**Automated 24-h Cattle Feeding Behaviour Analysis Using Raspberry Pi, Infrared Imaging, and Deep Learning-Based Head Position and ID Classification**	Wager-Jones, G., Butler, M., Harris, W. E., Rutter, M., Bleach, E. & Behrendt, K. (2026)	Biosystems Engineering	2026 (month not confirmed)	Low-cost Raspberry Pi with motion-triggered infrared cameras at the feed bunk; VGG16 (TensorFlow/Keras) models classify head position and animal ID; muzzle-to-feed events are matched to feed bin weight to estimate intake per event.	Works day and night under real farm conditions; non-invasive alternative to wearables; links behavior to individual animals and feed weight.	Training and inference were run offline, not in real time; needs large, varied training data; depends on bin weight data and camera placement.
3	**Multi-Object Detection of Forage Density and Dairy Cow Feeding Behavior Based on an Improved YOLOv10 Model for Smart Pasture Applications**	Liu, Z. et al. (2026)	Sensors, 26(4), 1273	Feb 2026	BFDet-YOLO: lightweight YOLOv10 variant with feature fusion, an occlusion-handling module (SEAM) and an improved DIoU box loss; jointly detects cow head/eating and forage density level on a custom pasture dataset.	Links feed availability with feeding behavior in one model; very small (1.85 M parameters, 4.45 MB) so suited to edge devices; precision 94.2%, recall 93.5%, mAP@50 95.7%.	Forage is graded by density level, not weighed in kg; mAP@50-95 is lower (70.7%); pasture-specific dataset with limited multi-farm validation.
4	**Modernizing Livestock Operations: Smart Feedlot Technologies and Their Impact**	Dao, S. D., Gostar, A. K., Tennakoon, R., Chuah, W. Q. & Bab-Hadiashar, A. (2026)	Animals, 16(8), 1244	Apr 2026	Narrative review of about 350 publications (117 cited) covering RFID, automated weighing, machine vision, smart troughs, feed/water intake monitoring and AI decision support in feedlots.	Covers the whole feedlot pipeline; links precision feeding to better feed use and less waste; sets out priorities such as edge-cloud design and multi-sensor fusion.	Review only (no new experiments); intake-monitoring systems are mostly validated in trials, not commercial multi-breed herds; cost, durability and data governance remain barriers.
5	**Cattle Farming Activity Monitoring Using Advanced Deep Learning Approach**	Asim, M. et al. (2026)	Sensors, 26(3), 785	Jan 2026	Comparison of YOLOv8 and YOLOv9 detectors for classifying fine-grained cattle activities from farm camera images.	YOLOv8-L reached 91.11% mAP (YOLOv9-E 90.23%); stable inference time; practical camera-based monitoring.	Behavior classification only (no feed quantity); small gap between YOLO variants; generalization across farms and breeds not shown.
6	**Integrating YOLOv8-pose for Localization and Behavior Tracking of Calves in Precision Livestock Farming**	Jahn, S., Düpjan, S., Röttgen, V., Jordt, J.-N., Albrecht, S. & Homeier-Bachmann, T. (2026)	Frontiers in Animal Science	Jan 2026	YOLOv8-pose keypoint model that tracks calf posture and location and recognizes feeding and drinking from video.	Fast video analysis; feeding periods matched manual annotation closely (about 99% overlap); also provides posture data.	Experimental setting only; could not tell apart two feeding positions at the same fence; calves only.
7	**Beef Cattle Behavior Recognition Based on Nighttime Farm Videos via Spatio-Temporal Enhancement and Dynamic Fusion**	Han, Y., Zhang, Z., Zhang, W., Cao, S., Sun, Y., Jia, Z., Wu, D., Huang, L. & Zhang, H. (2026)	Animals, 16(12), 1881	Jun 2026	STED-Net: spatio-temporal dark-video enhancement module plus dynamic fusion block; new dataset of 1,097 night clips from 30 Angus cattle with 6 behaviors including feeding.	Handles dark farm videos; precision 88.47%, recall 80.18%, accuracy 83.80%, F1 84.12%; feeding was among the best classes (F1 87.0%).	About 292 ms per sample, too slow for edge or strict real-time use; imbalanced classes and a single farm; no feed quantity.
8	**Advancing Standardized Cattle Behavior Classification with a Random Forest Model Evaluated Across Diverse Datasets**	James, H., Rial, C., Giordano, J. & Erickson, D. (2026)	Frontiers in Animal Science	Jan 2026	Random Forest on accelerometer features (nose, ear, neck sensors) across five datasets; classifies eating, rumination and movement.	One standard pipeline works across datasets; average 92.7% accuracy (80/20 split); low compute for battery-powered devices.	Machine learning, not deep learning; accuracy drops under stricter validation (e.g., 93.4% to 61.5% leave-one-individual-out); weak transfer between datasets; needs a sensor on each animal.
9	**A Cattle Behavior Recognition Method Based on Graph Neural Network Compression on the Edge**	Liu, H., Song, P., Xin, X., Rong, Y., Gao, J., Wang, Z. & Zhang, Y. (2026)	Animals, 16(3), 430	Jan 2026	S-ResNet for single-frame sensor input compressed with a graph neural network, run on a wearable low-power microcontroller.	On-device inference lowers latency and power use and extends battery life; suited to large free-grazing herds.	Needs a device on each animal; I could not retrieve the per-behavior results or whether feeding is classified separately.
10	**BoviFusionNet: A Lightweight Edge-Deployable AI System for Cattle Behavior Recognition in Livestock Monitoring**	Li, J., Zhang, W., Ren, X., He, J., Wang, L., Lv, J., Lin, K., Du, W. & Chen, R. (2026)	Open-access journal (title not confirmed)	Jul 2026	Lightweight object detector with box-balanced learning, better downsampling, multi-scale fusion and fine-grained discrimination; box-balanced dataset for eating, standing and lying.	Built for edge deployment on barns and pastures; targets dense, occluded cattle and visually similar behaviors.	Only three behaviors; I did not retrieve its accuracy figures; detects eating but not feed quantity.

Research Gap

Synthesizing the limitations reported across the ten reviewed studies (2026), the following gaps emerge in AI-based cattle feed and feeding monitoring:

- **Behavior detected, feed quantity not measured.** Most studies (Refs 5-10) recognize feeding or eating as a behavior but do not estimate how much feed was eaten or left. Only Refs 1-3 link behavior to feed weight, intake or forage density.
- **Individual-level intake is hard to obtain.** Ref 1 predicts intake well but relies on electronic feeders (GrowSafe); Ref 2 links feeding events to bin weight but works offline. Low-cost, real-time, per-animal intake from cameras is still unsolved.
- **Limited multimodal fusion.** Vision systems (Refs 2, 3, 5, 6, 7, 10) and sensor or wearable systems (Refs 1, 8, 9) are mostly studied separately; few combine camera, weight and accelerometer data.
- **Small, single-site datasets.** Studies use one feedlot (Ref 1), one pasture dataset (Ref 3), 30 Angus cattle (Ref 7) or experimental calf pens (Ref 6), which limits generalization across breeds, lighting and farm layouts.
- **Weak validation across animals and datasets.** Ref 8 shows accuracy falling from 93.4% to 61.5% when tested on unseen individuals, and transfer between datasets was often poor.
- **Day, night and outdoor robustness.** Only Refs 2 and 7 address night-time monitoring; dust, rain and occlusion at feed bunks and pastures remain hard.
- **Accuracy vs. edge efficiency.** Ref 7 takes about 292 ms per sample, too slow for edge use, while lightweight models (Refs 3, 9, 10) lack long-term field deployment results.
- **Wearable burden.** Sensor approaches (Refs 8, 9) need a device on every animal, raising cost and maintenance for large herds.
- **No unified benchmark and thin adoption evidence.** Datasets, behavior classes and metrics (mAP, accuracy, F1, RMSE) differ between studies, and few report cost, feed-waste savings or usability for smaller farms (Ref 4).

SDG Mapping

The reviewed literature aligns with the following United Nations Sustainable Development Goals (SDGs):

SDG	Relevant Studies	Mapping Rationale
SDG 2 - Zero Hunger	Refs 1-10 (all)	Better feed and feeding monitoring improves livestock productivity and the efficiency of milk and meat supply.
SDG 9 - Industry, Innovation and Infrastructure	Refs 1-4, 7, 9, 10	Edge AI, lightweight YOLO models, electronic feeding and smart-feedlot technology are innovations in agricultural infrastructure.
SDG 12 - Responsible Consumption and Production	Refs 1, 2, 3, 4	Measuring feed intake, leftovers and forage availability helps cut feed waste and supports precise feeding.
SDG 13 - Climate Action	Refs 1, 4	Precision feeding and climate-adaptive feed management can lower emissions per unit of beef or milk.
SDG 15 - Life on Land	Refs 3, 8	Pasture feed monitoring and behavior-based grazing management (e.g., virtual fencing) support sustainable use of grazing land.

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